

Joint Distributions

Bivariate distribution

Definition

X, Y : discrete random variables defined on the same sample space.

A, B : sets of possible values of X and Y .

$$p(x, y) = P(X = x, Y = y) = P(A \cap B)$$

$$\sum_{x \in A} \sum_{y \in B} p(x, y) = 1$$

Probability mass function of X

$$p(x) = P(X = x) = P(x = x, y \in B) = \sum_{y \in B} p(x, y)$$

Theorem

$p(x, y)$: Joint probability mass function of discrete R.V. X and Y .

$h(X, Y)$: A function of X and Y

$$E[h(X, Y)] = \sum_{x \in A} \sum_{y \in B} h(x, y)p(x, y)$$

$$E(X + Y) = E(X) + E(Y)$$

Proof

$$\begin{aligned} E[X + Y] &= \sum_{x \in A} \sum_{y \in B} (x + y)p(x, y) \\ &= \sum_{x \in A} \sum_{y \in B} xp(x, y) + \sum_{x \in A} \sum_{y \in B} yp(x, y) \\ &= \sum_{x \in A} x \left(\sum_{y \in B} p(x, y) \right) + \sum_{y \in B} y \left(\sum_{x \in A} p(x, y) \right) \\ &= \sum_{x \in A} xp_X(x) + \sum_{y \in B} yp_Y(y) = E[X] + E[Y] \end{aligned}$$

Definition

Two r. v.'s X and Y , defined on the same sample space, have a continuous joint distribution if there exists a nonnegative function of 2 variables, $f(x, y)$ on $\mathbb{R} \times \mathbb{R}$ such that for some region S in the xy -plane

$$P((X, Y) \in S) = \iint_S f(x, y) dx dy$$

$f(x, y)$: is called the joint (probability) density function of X and Y .

Definition

單點機率為 0 : $P(X = a, Y = b) = \int_b^b \left(\int_a^a f(x, y) dx \right) dy = 0$

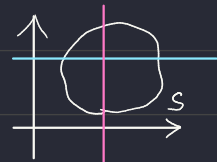
總和機率為 1 : $A = (-\infty, +\infty), B = (-\infty, +\infty) \implies \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$

Definition

Let X and Y have joint density function $f(x, y)$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$



are called, the marginal (probability) density functions of X and Y .

Definition

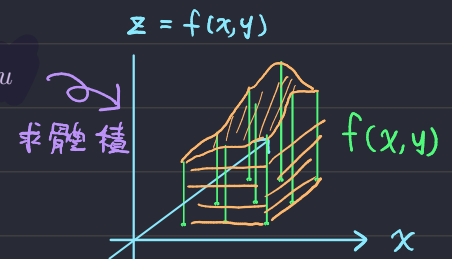
Joint (cumulative) distribution function

$$F(X, Y) = F(t, u) = P(X \leq t, Y \leq u) = \int_{-\infty}^u \int_{-\infty}^t f(u, t) dy du$$

The marginal distribution function

$$F_X(t) = P(X \leq t) = P(X \leq t, Y \leq \infty) = F(t, \infty)$$

$$F_Y(t) = P(Y \leq t) = P(X \leq \infty, Y \leq t) = F(\infty, t)$$

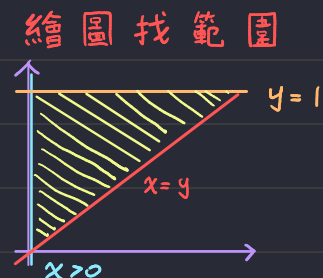


Ex 8.4 The joint density function of random variables X and Y is given by

$$f(x, y) = \begin{cases} \lambda xy^2 & 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Determine the value of λ

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \int_0^1 \int_0^y \lambda xy^2 dx dy \\ &= \int_0^1 \lambda \left(\frac{1}{2} x^2 y^2 \right) \Big|_0^y dy \\ &= \lambda \int_0^1 \frac{1}{2} y^4 dy \\ &= \lambda \left(\frac{1}{10} y^5 \right) \Big|_0^1 \\ &= \frac{1}{10} \lambda \end{aligned}$$

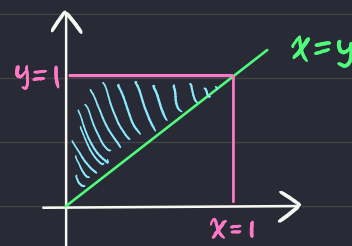


Ans: $\lambda = 10$ #

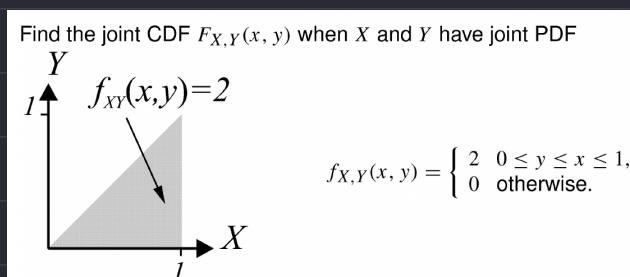
(b) Find the marginal density function of X and Y .

作答注意範圍

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y) dy \\ &= \int_x^1 f(x,y) dy \\ &= 10 \left(\frac{1}{3} x y^3 \right) \Big|_x^1 \\ &= \frac{10}{3} (x - x^4) \quad \# \quad \text{if } 0 \leq x \leq 1 \end{aligned}$$



$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f(x,y) dx \\ &= \int_0^y 10 x y^2 dx \\ &= 5 x^2 y^2 \Big|_0^y \\ &= 5 y^4 \quad \# \quad \text{if } 0 \leq y \leq 1 \end{aligned}$$



$$\begin{aligned} F(t,u) &= P(X \leq t, Y \leq u) \\ &= \int_{-\infty}^t \int_{-\infty}^u f(x,y) dt du \\ &= \int_0^t \int_u^x 2 dt du \\ &= \int_0^t (2t) \Big|_u du \\ &= \int_0^t 2x - 2u du \\ &= (2xu - u^2) \Big|_0^t \\ &= 2xt - t^2 \end{aligned}$$

後面還有一坨範圍判定



Ex 8.5

For $\lambda > 0$, let

$$F(x, y) = \begin{cases} 1 - \lambda e^{-\lambda(x+y)} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Determine if F is the joint distribution function of 2 random variables X and Y.

$$\frac{\partial^2}{\partial x \partial y} F(x, y)$$

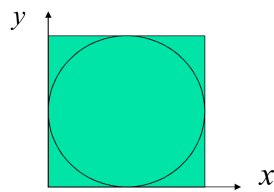
$$= \frac{\partial}{\partial x} -\lambda e^{-\lambda(x+y)} \cdot (-\lambda)$$

$$= \frac{\partial}{\partial x} \lambda^2 e^{-\lambda(x+y)}$$

$$= -\lambda^3 e^{-\lambda(x+y)} \quad \text{if } x > 0, y > 0$$

↳ 負的，所以不行 ☹️ #

Ex 8.6 A circle of radius 1 is inscribed in a square with sides of length 2. A point is selected at random from the square. What is the probability that the point is inside the circle?



$\frac{\pi}{4}$ #

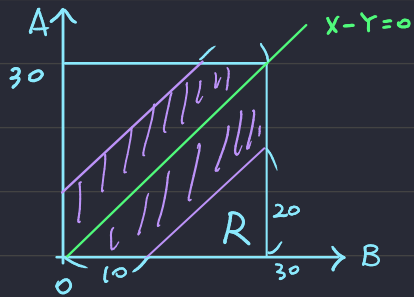
Definition

Let S be a subset of the plane with area $A(S)$

A point is said to be randomly selected from S if for any subset R of S with area $A(R)$, the probability that R contains the point is $A(R)/A(S)$.

Ex 8.7 A man invites his fiancée to a fine hotel for a Sunday brunch. They decide to meet in the lobby of the hotel between 11:30 A.M. and 12 noon. If they arrive at random times during this period, what is the probability that they will meet within 10 minutes?

面積法:



R : possible meet time

$S: |X - Y| \leq 10$

$$A: 30 \times 30 - 2 \times (20 \times 20 \times \frac{1}{2}) = 500$$

$$\text{ANS: } \frac{500}{900}$$

Theorem

$Z = h(x, y)$: The actual value of the function

$f(x, y)$: The p.d.f. of continuous R.V. X and Y

$$E(Z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy$$

$$E(X + Y) = E(X) + E(Y)$$

Proof

$$\begin{aligned} E(X + Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy \\ &= E(X) + E(Y). \end{aligned}$$

Theorem

$$E(Z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy$$

$$Z = h(x, y)$$

$f(x, y)$: The probability density function

Corollary

For continuous r. v.'s X and Y , $E(X + Y) = E(X) + E(Y)$

$$\begin{aligned} E(X + Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f(x, y) dx dy && \text{marginal density function} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy \\ &= E(X) + E(Y). \end{aligned}$$

Ex 8.9 Let X and Y have joint density function

$$f(x, y) = \begin{cases} \frac{3}{2}(x^2 + y^2) & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X^2 + Y^2)$

$$h(x, y) = x^2 + y^2$$

$$\begin{aligned} E(X^2 + Y^2) &= E(h(x, y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) \cdot f(x, y) dx dy \\ &= \int_0^1 \int_0^1 (x^2 + y^2) \cdot \frac{3}{2} (x^2 + y^2) dx dy \\ &= \frac{3}{2} \int_0^1 \int_0^1 x^4 + 2x^2y^2 + y^4 dx dy \\ &= \frac{3}{2} \int_0^1 \left(\frac{1}{5}x^5 + \frac{2}{3}x^3y^2 + y^4x \right) \Big|_0^1 dy \\ &= \frac{3}{2} \int_0^1 \left(\frac{1}{5} + \frac{2}{3}y^2 + y^4 \right) dy \\ &= \frac{3}{2} \left(\frac{1}{5}y + \frac{2}{9}y^3 + \frac{1}{5}y^5 \right) \Big|_0^1 = \frac{3}{2} \cdot \frac{28}{45} = \frac{14}{15} \end{aligned}$$

8.2 Independent random variables

Definition

A, B are called independent if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$

If A, B are independent, then :

Probability $P(X \leq a, Y \leq b) = P(X \leq a)P(Y \leq b)$

Joint distribution function $F(t, u) = P(X \leq t, Y \leq u) = F_X(t)F_Y(u)$

Joint PMF $p(x, y) = p_X(x)p_Y(y)$ 離散

Joint density function $f(x, y) = f_X(x)f_Y(y)$ 連續

Theorem 8.5

X, Y: two independent R.V.

g, h: two $\mathbb{R} \rightarrow \mathbb{R}$ function

$\rightarrow g(X)$ and $h(Y)$ are also independent

Theorem 8.6

X, Y: two independent R.V

g, h: two $\mathbb{R} \rightarrow \mathbb{R}$ function

$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$

本理論不是雙向的，如果

$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$

成立，不代表他們一定獨立

X: a uniform distribution in $[-1, 1]$

$Y = X^2$

$E(X) = 0$ $E(X) \cdot E(Y) = 0$

$E(XY) = E(X \cdot X^2) = E(X^3) = 0$

Proof $E(Z) = E[g(X)h(Y)]$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [g(x)h(y)] \cdot f(x, y) \, dx \, dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [g(x)h(y)] \cdot [f_X(x) \cdot f_Y(y)] \, dx \, dy$$

$$= \int_{-\infty}^{\infty} h(y) f_Y(y) \, dy \cdot \int_{-\infty}^{\infty} g(x) f_X(x) \, dx$$

$$= E[h(Y)] \cdot E[g(X)]$$

Ex 8.12 Store A and B, which belong to the same owner, are located in 2 different towns. If the density function of the weekly profit of each store, in thousands of dollars, is given by

$$f(x) = \begin{cases} x/4 & \text{if } 1 < x < 3 \\ 0 & \text{otherwise,} \end{cases}$$

and the profit of one store is independent of the other, what is the probability that next week one store makes at least \$500 more than the other store?

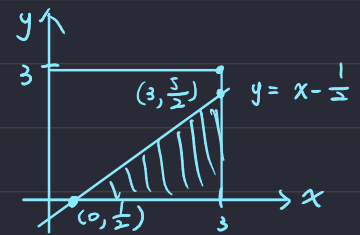
$$P(X > Y + \frac{1}{2}) + P(Y > X + \frac{1}{2}) = 2P(X > Y + \frac{1}{2})$$

$$f_x(x) = \begin{cases} \frac{1}{4}x & \text{if } 1 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

$$f_y(y) = \begin{cases} \frac{1}{4} & \text{if } 1 < y < 3 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x,y) = \begin{cases} \frac{1}{16}xy & \text{if } 1 < x < 3, 1 < y < 3 \\ 0 & \text{otherwise} \end{cases}$$

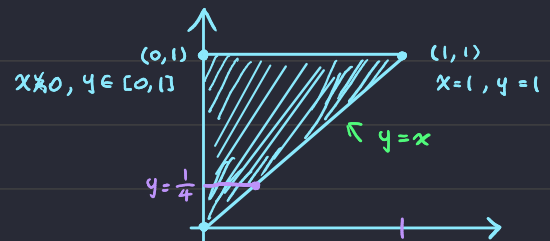
$$\begin{aligned} & \int_{\frac{1}{2}}^3 \int_0^{x-\frac{1}{2}} f(x,y) dy dx \\ &= \int_{\frac{1}{2}}^3 \int_0^{x-\frac{1}{2}} \frac{1}{16} xy dy dx \\ &= \dots \text{ (計算過程略)} \\ &= \frac{549}{1024} \end{aligned}$$



Ex 8.15 Prove that 2 random variables X and Y with the following joint density function are not independent.

$$f(x,y) = \begin{cases} 8xy & 0 \leq x \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

$$\begin{aligned} f_x(x) &= \int_x^1 8xy dy \\ &= 4xy^2 \Big|_x^1 = 4x - 4x^3 \\ f_y(y) &= \int_0^y 8xy dx \\ &= 4x^2y \Big|_0^y = 4y^3 \end{aligned}$$



找 x 範圍

1. 找隨意 y 值
2. 看 x 的 Range
3. 0 ~ y

$$f(x,y) \neq f_x(x) \cdot f_y(y) \neq$$

8.3 Conditional distributions

Definition

The conditional probability mass function of X given that $Y=y$

$$\begin{aligned} p_{X|Y}(X|y) &= p_X(X, Y) = P(X|Y) = P(X=x | Y=y) \\ &= \frac{P(X=x, Y=y)}{P(Y=y)} = \frac{p(x, y)}{p_Y(y)} \end{aligned}$$

$$\sum_{x \in A} = \sum_{x \in A} \frac{p(x, y)}{p_Y(y)} = 1$$

$$F_{X|Y}(x|y) = P(X \leq x | Y = y) = \sum_{t \leq x} P(X = t, Y = y) = \sum_{t \leq x} p_{X|Y}(t|y)$$



微分

$$f_{X|Y}(x, y) = \frac{f(x, y)}{f_Y(y)}$$

If X and Y are two independent variable

$$p_{X|Y}(x|y) = p(x)$$

$$f_{X|Y}(x|y) = f_X(x)$$

Ex 8.21 Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} \frac{3}{2}(x^2 + y^2) & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $f_{X|Y}(x|y)$.

$$f(x|y) = \frac{f(x, y)}{f(y)}$$

$$\begin{aligned} f(y) &= \int_0^1 \frac{3}{2}(x^2 + y^2) dx \\ &= \frac{3}{2}y^2 + \left(\frac{1}{2}x^3\right) \Big|_0^1 \\ &= \frac{3}{2}y^2 + \frac{1}{2} \end{aligned}$$

$$f_{X|Y}(x|y) = \frac{3/2(x^2 + y^2)}{(3/2)y^2 + 1/2} = \frac{3(x^2 + y^2)}{3y^2 + 1}$$

for $0 < x < 1$ and $0 < y < 1$.

Everywhere else, $f_{X|Y}(x|y) = 0$.

注意
範圍



Ex 8.22 First, a point Y is selected at random from the interval $(0,1)$. Then another point X is chosen from the interval $(0,Y)$. Find the density function of X .

$$f(y) = \begin{cases} 1 & \text{if } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x|y) = \begin{cases} \frac{1}{y} & \text{if } 0 \leq x \leq y, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

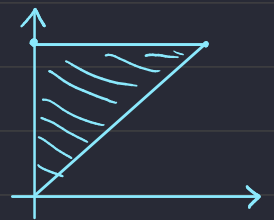
$$f(x|y) = \frac{f(x, y)}{f(y)}$$

$$f(x, y) = f(x|y) \cdot f(y)$$

$$= \frac{1}{y} \text{ if } 0 \leq x \leq y \leq 1$$

$$f(x) = \int_x^1 \frac{1}{y} dy$$

$$= -\ln(x) \text{ if } 0 \leq x \leq 1, \text{ elsewhere } 0$$



Ex 8.18 Calculate the expected number of aces in a randomly selected poker hand that is found to have exactly 2 jacks.

5 cards

X : The number of ace

Y : The number of jack

$$E(X|Y=2) = \sum_{x=0}^3 x \cdot p(x|Y=2) \rightarrow \boxed{J} \boxed{J} \overbrace{\boxed{?} \boxed{?} \boxed{?}}^3$$

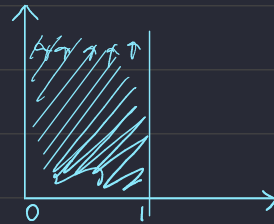
$$= \sum_{x=0}^3 x \cdot \frac{C_4^x \cdot C_2^4 \cdot C_{5-x-2}^{44} / C_5^{52}}{C_2^4 \cdot C_3^{48} / C_5^{52}}$$

$$\approx 0.25 \#$$

Ex 8.24 Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} e^{-y} & \text{if } y > 0, 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X|Y=2)$.



$$E(X|Y=2)$$

$$= \int_{-\infty}^{\infty} x \cdot f(x, 2) dx$$

$$f(x, 2) = \frac{f(x, 2)}{f_Y(2)}$$

$$f_Y(2) = \int_{-\infty}^{\infty} f(x, 2) dx$$

$$= \int_0^1 e^{-2} dx = e^{-2}$$